Hao Li, Ph.D.

Applied Scientist, AWS
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Current Position

Amazon Web Services — Seattle, WA
Applied Scientist II, AWS Rekognition — Oct 2018 - present

- Research and develop machine learning and computer vision algorithms for AWS image and video analytical services, contributed to the launch of *Custom Labels*, *Content Moderation* and *Lookout for Vision*. Research on efficient and automatic ML for transfer learning, hyperparameter optimization and neural architecture search.

Research Interests

My research interests lie at the intersection of machine learning, computer vision, and distributed computing, with the goal of making machine learning algorithms and their applications more efficient, scalable, explainable, automatic, and transferable.

Education

University of Maryland, College Park — College Park, MD
Ph.D. in Computer Science — 2012 – 2018

- Dissertation: Towards Fast and Efficient Representation Learning
- Advisors: Prof. Hanan Samet and Prof. Tom Goldstein
- Committee: Hanan Samet, Tom Goldstein, Larry Davis, David Jacobs, Rama Chellappa

Institute of Computing Technology, Chinese Academy of Sciences — Beijing, China
M.E. in Computer Science and Technology — 2009 – 2012

- Thesis: Multi-Person Tracking with Re-identification in Non-overlapping Camera Networks
- Advisor: Prof. Shiguang Shan

Shandong University — Jinan, China
B.E. in Software Engineering — 2005 – 2009

Industry Experience

AWS Rekognition, Amazon Web Services — Seattle, WA
Applied Scientist II — Oct 2018 - present

Machine Learning Team, Nervana Systems (now Intel AI) — San Diego, CA
Algorithms Engineer Intern with Scott Cyphers and Arjun Bansal — Summer 2016

Machine Learning Department, NEC Labs America — Princeton, NJ
Research Intern / Visiting Student — Summer 2015/Spring 2016

Storage System Department, NEC Labs America — Princeton, NJ
Research Intern with Asim Kadav and Erik Kruus — Summer 2014

Internet Graphics Group, Microsoft Research Asia — Beijing, China
Research Intern with Ying-Qing Xu — May – Nov 2010

Media Computing Group, Microsoft Research Asia — Beijing, China
Research Intern with Meng Wang — Mar – July 2009

Publications (Google Scholar, total citations: **2,965**)

Journal

- *Multimodal Graph-Based Reranking for Web Image Search*
  Meng Wang, **Hao Li**, Dacheng Tao, Ke Lu, Xindong Wu
  IEEE Transactions on Image Processing (**T-IP**), 2012 (citations: **332**)

Pre-prints

- *A linearized framework and a new benchmark for model selection for fine-tuning*
  Aditya Deshpande, Alessandro Achille, Avinash Ravichandran, **Hao Li**, Luca Zancato, Charless Fowlkes, Rahul Bhotika, Stefano Soatto, Pietro Perona
  arXiv preprint arXiv:2102.00084, 2021

Conference & Workshop

- *Rethinking the Hyperparameters for Fine-tuning*
  Hao Li, Pratik Chaudhari, Hao Yang, Michael Lam, Avinash Ravichandran, Rahul Bhotika, Stefano Soatto, International Conference on Learning Representations (**ICLR**), 2020 (citations: 17)

- *Visualizing the Loss Landscape of Neural Nets*
  Hao Li, Zheng Xu, Gavin Taylor, Tom Goldstein
  Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2018 (citations: **566**)

- *Training Quantized Nets: A Deeper Understanding*
  Hao Li*, Soham De*, Zheng Xu, Christoph Studer, Hanan Samet, Tom Goldstein
  Annual Conference on Neural Information Processing Systems (**NeurIPS**), 2017 (citations: **115**)
  ICML Workshop on Principled Approaches to Deep Learning, 2017 (**Google Best Student Paper**)

- *Pruning Filters for Efficient ConvNets*
  Hao Li, Asim Kadav, Igor Durdanovic, Hanan Samet, Hans Peter Graf
  International Conference on Learning Representations (**ICLR**), 2017 (citations: **1,717**)
  NIPS Workshop on Efficient Methods for Deep Neural Networks (**EMDNN**), 2016

- *Adaptive Consensus ADMM for Distributed Optimization*
  Zheng Xu, Gavin Taylor, **Hao Li**, Mario Figueiredo, Xiaoming Yuan, and Tom Goldstein
  International Conference on Machine Learning (**ICML**), 2017 (citations: **37**)

- *MALT: Distributed Data-Parallelism for Existing ML Applications*
  Hao Li, Asim Kadav, Erik Kruus, Cristian Ungureanu
  ACM European Conference on Computer Systems (**EuroSys**), 2015 (citations: **53**)
  NIPS Workshop on Machine Learning Systems (**LearningSys**), 2015

- *Streaming News Image Summarization*
  Hao Li, Shangfu Peng, Hanan Samet
  International Conference on Pattern Recognition (**ICPR**), 2016 (Oral)

- *Optimizing Multimodal Reranking for Web Image Search*
  Hao Li, Meng Wang, Zhisheng Li, Zheng-Jun Zha, Jialie Shen
  ACM SIGIR Conf. on Research and Development in Information Retrieval (**SIGIR**), 2011 (Short)

- *MSRA-MM 2.0: A Large-Scale Web Multimedia Dataset*
  Hao Li, Meng Wang, Xian-Sheng Hua
  IEEE ICDM Workshop on Internet Multimedia Mining, 2009 (citations: **102**)

- *Simplification and Refinement for Speedy Spatio-temporal Hot Spot Detection Using Spark (GIS Cup).*
  Shangfu Peng, Hong Wei, **Hao Li**, and Hanan Samet
  ACM Int. Conf. on Advances in Geographic Information Systems (**SIGSPATIAL**), 2016
- *View Adaptive Metric Learning for Multi-view Person Re-identification*
  Canxiang Yang, Shiguang Shan, Dan Wang, **Hao Li**, Xilin Chen
  Asian Conference on Computer Vision (**ACCV**), 2014

Patents

Passive Pruning of Filters in a Convolutional Neural Network. US Patent. 10,832,136

Parallelized Machine Learning with Distributed Lockless Training. US Patent 9,984,337

Security System Using a Conv. Neural Network with Pruned Filters. US Patent 10,885,437

Awards

Google Best Student Paper Award, ICML 2017 PADL Workshop	2017
Dean's Fellowship, Department of Computer Science, University of Maryland	2012
Stars of Tomorrow Excellent Intern Award, Microsoft Research Asia	2010
Outstanding College Graduate, Shandong Province and Shandong University	2009
Meritorious Winner in Mathematical Contest in Modeling, USA	2008
National Second Prize in China Undergraduate Mathematical Contest in Modeling	2007

Invited Talks

Rethinking the Hyperparameters for Fine-tuning

- Amazon Computer Vision Conference (ACVC), June 2020

Towards Fast and Efficient Representation Learning

- Facebook Research, Menlo Park, CA, May 2018
- IBM Research, Cambridge, MA, May 2018
- Amazon Web Services, Seattle, WA, April 2018
- Apple AI Research, Cupertino, CA, April 2018
- Vicarious, Union City, CA, April, 2018
- Comcast Labs, Washington D.C., Mar. 2018
- Microsoft Research Asia, Bejing, Jan. 2018
- NVIDIA Research, Santa Clara, CA, Nov. 2017

Towards a Deeper Understanding of Training Quantized Neural Networks

- ICML Workshop on Principled Approaches on Deep Learning, Sydney, Aug. 2017

Streaming News Image Summarization.

- ICPR, Cancun, Mexico, Dec. 2016

MALT: Distributed Data-Parallelism for Existing ML Applications.

- EuroSys, Bordeaux, France, April 2015

Teaching Experience

TA, CMSC 764 Advanced Numerical Optimization, with Prof. Tom Goldstein — Spring 2017
TA, CMSC 828Z Reinforcement Learning, with Prof. Hal Daumé III — Fall 2016
TA, CMSC 250H Discrete Structures (Honors), with Prof. Tom Goldstein — Fall 2017
TA, CMSC 216 Introduction to Computer Systems, with Dr. Larry Herman — Fall 2015
TA, CMSC 426 Programming Handheld Systems, with Prof. Atif Memon — Spring 2013
TA, CMSC 330 The Organization of Programming Languages, with Dr. Larry Herman — Fall 2012

Academic Services

Journal Reviewer

Journal of Machine Learning Research (JMLR) (**Editorial board reviewer**)
International Journal of Computer Vision (IJCV)
IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)
IEEE Transactions on Neural Network and Learning Systems (T-NNLS)
IEEE Transactions on Multimedia (T-MM)
IEEE Transactions on Industrial Informatics
Neurocomputing
Multimedia Systems

Conference Reviewer

Neural Information Processing Systems (NeurIPS)
International Conference on Machine Learning (ICML)
International Conference on Learning Representation (ICLR)
IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
International Conference on Computer Vision (ICCV)
European Conference on Computer Vision (ECCV)
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)
AAAI Conference on Artificial Intelligence (AAAI)